

```

<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Sistema Logístico - El Salvador</title>

  <!-- Fuentes Google -->
  <link rel="preconnect" href="https://fonts.googleapis.com" />
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
  <link href="https://fonts.googleapis.com/css2?family=Space+Grotesk:wght@400;600;700&family=Inter:wght@400;500;600;700&display=swap" rel="stylesheet" />

  <!-- MapLibre GL CSS y JS -->
  <link href="https://unpkg.com/maplibre-gl@3.6.2/dist/maplibre-gl.css" rel="stylesheet" />
  <script src="https://unpkg.com/maplibre-gl@3.6.2/dist/maplibre-gl.js"></script>

<style>
:root {
  --noche-900: #081826;
  --noche-800: #0b1a2b;
  --noche-700: #122a44;
  --papel-000: #f7f8fa;
  --papel-100: #eef0f3;
  --linea-borde: #e2e5ea;
  --tinta-900: #16202b;
  --tinta-500: #5b6472;
  --tinta-300: #9aa3af;
  --teal: #2ec4b6;
  --amber: #f5a623;
  --rojo: #ef4b4b;
  --verde: #38a169;
  --radio-s: 8px;
  --radio-m: 12px;
  --fuente-display: "Space Grotesk", "Inter", system-ui, sans-serif;
  --fuente: "Inter", system-ui, sans-serif;
}

* { margin: 0; padding: 0; box-sizing: border-box; }

body {
  font-family: var(--fuente);
  background: var(--papel-000);
  color: var(--tinta-900);
  -webkit-font-smoothing: antialiased;
  height: 100vh;
  overflow: hidden;
}

```

```
h1, h2, h3, .font-display { font-family: var(--fuente-display); }
```

```
button { font-family: inherit; cursor: pointer; }
```

```
input, select, textarea { font-family: inherit; }
```

```
input[type="text"], input[type="password"], input[type="email"], select {  
  border: 1px solid var(--linea-borde);  
  border-radius: var(--radio-s);  
  padding: 10px 12px;  
  font-size: 14px;  
  background: #fff;  
  color: var(--tinta-900);  
  width: 100%;  
}
```

```
input:focus-visible, select:focus-visible {  
  outline: none;  
  border-color: var(--amber);  
  box-shadow: 0 0 0 3px rgba(245, 166, 35, .18);  
}
```

```
/* ===== LOGIN
```

```
===== */
```

```
.pantalla {  
  position: fixed;  
  inset: 0;  
  z-index: 1000;  
  min-height: 100dvh;  
  display: grid;  
  grid-template-columns: 1.05fr 1fr;  
  background: var(--papel-000);  
}
```

```
.panel-marca {  
  background: linear-gradient(180deg, var(--noche-900) 0%, var(--noche-800) 100%);  
  color: #eef3f6;  
  padding: 56px 56px 40px;  
  display: flex;  
  flex-direction: column;  
  position: relative;  
  overflow: hidden;  
}
```

```
.marca {  
  display: flex;  
  align-items: center;  
  gap: 10px;  
  margin-bottom: 64px;
```

```
}
```

```
.marca-icono {  
  display: inline-flex;  
  width: 36px;  
  height: 36px;  
  border-radius: 8px;  
  background: rgba(255, 255, 255, 0.08);  
  align-items: center;  
  justify-content: center;  
}
```

```
.marca-texto {  
  font-family: var(--fuente-display);  
  font-weight: 600;  
  font-size: 18px;  
  letter-spacing: -0.01em;  
}
```

```
.titular {  
  font-size: clamp(32px, 3.4vw, 44px);  
  line-height: 1.12;  
  font-weight: 600;  
  color: #fff;  
  max-width: 12ch;  
}
```

```
.subtitular {  
  margin-top: 20px;  
  max-width: 38ch;  
  color: #aab8c4;  
  font-size: 15px;  
  line-height: 1.6;  
}
```

```
.pie-marca {  
  margin-top: auto;  
  font-size: 12.5px;  
  color: #7c8ba0;  
}
```

```
.panel-formulario {  
  background: var(--papel-000);  
  display: flex;  
  align-items: center;  
  justify-content: center;  
  padding: 40px;  
}
```

```
.tarjeta-login {  
  width: 100%;  
  max-width: 400px;  
}
```

```
.selector-rol {  
  display: flex;  
  background: var(--papel-100);  
  border-radius: var(--radio-m);  
  padding: 4px;  
  gap: 4px;  
  margin-bottom: 28px;  
}
```

```
.rol-btn {  
  flex: 1;  
  border: none;  
  background: transparent;  
  padding: 9px 6px;  
  border-radius: 7px;  
  font-size: 13px;  
  font-weight: 500;  
  color: var(--tinta-500);  
  cursor: pointer;  
  transition: background .15s, color .15s;  
}
```

```
.rol-btn.activo {  
  background: var(--papel-000);  
  color: var(--noche-900);  
  box-shadow: 0 1px 3px rgba(11, 26, 43, .12);  
  font-weight: 600;  
}
```

```
.form-titulo { font-size: 24px; font-weight: 600; }  
.form-subtitulo { margin-top: 6px; color: var(--tinta-500); font-size: 14px; }  
.form-login { margin-top: 24px; display: flex; flex-direction: column; gap: 16px; }
```

```
.campo { display: flex; flex-direction: column; gap: 6px; font-size: 13px; color: var(--tinta-500); }  
.campo span { font-weight: 500; }
```

```
.btn-entrar {  
  margin-top: 6px;  
  background: var(--amber);  
  color: #1a1204;  
  border: none;  
  border-radius: var(--radio-s);  
  padding: 13px;  
  font-size: 15px;
```

```
font-weight: 600;
transition: background .15s;
width: 100%;
}
```

```
.btn-entrar:hover { background: #e09117; }
```

```
/* ===== LAYOUT
PRINCIPAL ===== */
```

```
#app-screen {
  display: none;
  height: 100vh;
  flex-direction: column;
}
```

```
header {
  background: var(--noche-900);
  color: #fff;
  padding: 14px 28px;
  display: flex;
  justify-content: space-between;
  align-items: center;
  border-bottom: 1px solid rgba(255, 255, 255, 0.08);
  flex-shrink: 0;
}
```

```
.header-branding {
  display: flex;
  align-items: center;
  gap: 12px;
}
```

```
.header-title {
  font-family: var(--fuente-display);
  font-size: 17px;
  font-weight: 600;
  color: #fff;
}
```

```
.user-info {
  display: flex;
  align-items: center;
  gap: 14px;
}
```

```
.rol-etiqueta {
  font-size: 12px;
  font-weight: 600;
  text-transform: uppercase;
}
```

```
letter-spacing: .04em;
color: var(--teal);
border: 1px solid rgba(46, 196, 182, .4);
background: rgba(46, 196, 182, .08);
border-radius: 999px;
padding: 4px 12px;
}
```

```
.btn-salir {
border: 1px solid rgba(255, 255, 255, 0.2);
background: transparent;
color: #eef3f6;
border-radius: var(--radio-s);
padding: 6px 14px;
font-size: 13px;
font-weight: 500;
transition: background .15s;
}
```

```
.btn-salir:hover { background: rgba(255, 255, 255, 0.1); }
```

```
.main-container {
display: flex;
flex: 1;
height: calc(100vh - 59px);
overflow: hidden;
}
```

```
/* NAVEGACIÓN LATERAL */
```

```
.sidebar-nav {
width: 210px;
min-width: 210px;
background: var(--noche-800);
color: #fff;
display: flex;
flex-direction: column;
padding: 16px 10px;
gap: 6px;
border-right: 1px solid var(--noche-700);
}
```

```
.sidebar-nav button {
background: transparent;
border: none;
color: #aab8c4;
padding: 11px 14px;
text-align: left;
font-size: 13.5px;
font-weight: 500;
```

```

    cursor: pointer;
    border-radius: 8px;
    transition: background .15s, color .15s;
    display: flex;
    align-items: center;
    gap: 10px;
}

.sidebar-nav button:hover, .sidebar-nav button.active {
    background: var(--noche-700);
    color: #fff;
}

.sidebar-nav button.active {
    font-weight: 600;
    border-left: 3.5px solid var(--amber);
    border-radius: 4px 8px 8px 4px;
}

/* PANELES DE FORMULARIOS */
.panel-form {
    width: 380px;
    min-width: 380px;
    background: var(--papel-000);
    padding: 20px;
    border-right: 1px solid var(--linea-borde);
    overflow-y: auto;
    display: flex;
    flex-direction: column;
    gap: 18px;
}

.tarjeta {
    background: #fff;
    border: 1px solid var(--linea-borde);
    border-radius: var(--radio-m);
    box-shadow: 0 1px 2px rgba(11, 26, 43, .05);
    padding: 20px;
}

.tarjeta h2, .tarjeta h3 {
    font-size: 16px;
    font-weight: 600;
    color: var(--noche-900);
    margin-bottom: 4px;
}

.tarjeta p.sub {
    font-size: 13px;

```

```
color: var(--tinta-500);
margin-bottom: 14px;
}
```

```
.btn {
border: 1px solid var(--linea-borde);
background: transparent;
color: var(--tinta-500);
border-radius: var(--radio-s);
padding: 9px 14px;
font-size: 13px;
font-weight: 500;
transition: background .15s, border-color .15s;
cursor: pointer;
text-align: center;
}
```

```
.btn:hover { background: var(--papel-100); }
```

```
.btn-primario {
background: var(--noche-800);
color: #fff;
border-color: var(--noche-800);
width: 100%;
font-weight: 600;
}
```

```
.btn-primario:hover { background: var(--noche-900); }
```

```
.btn-ambar {
background: var(--amber);
color: #201303;
border-color: var(--amber);
font-weight: 600;
width: 100%;
}
```

```
.btn-ambar:hover { background: #e09117; }
```

```
.btn-secundario {
background: transparent;
color: var(--tinta-500);
border: 1px solid var(--linea-borde);
width: 100%;
margin-top: 8px;
}
```

```
/* ÁREA DE CONTENIDO (MAPA Y TABLA) */
```

```
.content-area {
flex: 1;
display: flex;
```

```
flex-direction: column;
height: 100%;
overflow: hidden;
background: var(--papel-000);
}
```

```
#map {
  flex: 1.2;
  width: 100%;
  position: relative;
}
```

```
.table-container {
  flex: 1;
  background: #fff;
  padding: 20px 24px;
  overflow-y: auto;
  border-top: 1px solid var(--linea-borde);
}
```

```
.table-top {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 14px;
}
```

```
.table-title {
  font-size: 16px;
  font-weight: 600;
  color: var(--noche-900);
}
```

```
.tabla {
  width: 100%;
  font-size: 13.5px;
  border-collapse: collapse;
}
```

```
.tabla th {
  text-align: left;
  border-bottom: 1px solid var(--linea-borde);
  padding: 10px 12px;
  font-size: 11.5px;
  font-weight: 600;
  text-transform: uppercase;
  letter-spacing: .04em;
  color: var(--tinta-300);
  background: #fff;
}
```

```
position: sticky;
top: 0;
z-index: 5;
}
```

```
.tabla td {
padding: 12px;
border-bottom: 1px solid var(--linea-borde);
vertical-align: middle;
}
```

```
.tabla tr:last-child td { border-bottom: none; }
```

```
/* BADGES Y PILLS */
```

```
.badge {
display: inline-flex;
align-items: center;
border-radius: 999px;
padding: 3px 10px;
font-size: 11.5px;
font-weight: 600;
}
```

```
.badge-PENDIENTE { background: rgba(245, 166, 35, .15); color: #9a6410; border: 1px solid
rgba(245, 166, 35, .4); }
```

```
.badge-ASIGNADO { background: rgba(128, 90, 213, .15); color: #6b46c1; border: 1px solid
rgba(128, 90, 213, .4); }
```

```
.badge-EN_CAMINO { background: rgba(49, 130, 206, .15); color: #2b6cb0; border: 1px solid
rgba(49, 130, 206, .4); }
```

```
.badge-ENTREGADO { background: rgba(46, 196, 182, .15); color: #0b8378; border: 1px solid
rgba(46, 196, 182, .4); }
```

```
.badge-INCIDENCIA { background: rgba(239, 75, 75, .15); color: #c23333; border: 1px solid
rgba(239, 75, 75, .4); }
```

```
.badge-CANCELADO { background: rgba(113, 128, 150, .15); color: #4a5568; border: 1px solid
rgba(113, 128, 150, .4); }
```

```
.btn-action {
padding: 4px 10px;
font-size: 12px;
border-radius: 6px;
border: 1px solid var(--linea-borde);
background: #fff;
cursor: pointer;
margin-right: 4px;
transition: background .15s;
}
```

```
.btn-action:hover { background: var(--papel-100); }
```

```
.btn-edit { color: var(--noche-800); font-weight: 500; }
```

```
.btn-del { color: var(--rojo); border-color: rgba(239, 75, 75, .3); }
.btn-del:hover { background: rgba(239, 75, 75, .08); }
```

```
.oculto { display: none !important; }
```

```
@media (max-width: 900px) {
  .pantalla { grid-template-columns: 1fr; }
  .panel-marca { display: none; }
  .main-container { flex-direction: column; }
  .sidebar-nav { width: 100%; flex-direction: row; overflow-x: auto; }
  .panel-form { width: 100%; }
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<!-- ===== VISTA LOGIN ===== -->
```

```
<main id="login-screen" class="pantalla">
```

```
<section class="panel-marca">
```

```
<div class="marca">
```

```
<span class="marca-icono" aria-hidden="true">
```

```
<svg viewBox="0 0 32 32" width="22" height="22">
```

```
<path d="M4 20 C 9 8, 18 22, 28 6" fill="none" stroke="#2ec4b6" stroke-width="2.6" stroke-
linecap="round"/>
```

```
<circle cx="4" cy="20" r="3" fill="#f5a623"/>
```

```
<circle cx="28" cy="6" r="3" fill="#f5a623"/>
```

```
</svg>
```

```
</span>
```

```
<span class="marca-texto">Transporte & Entregas</span>
```

```
</div>
```

```
<h1 class="titular">Cada pedido,<br />visible en su ruta.</h1>
```

```
<p class="subtitular">
```

```
Administradores, conductores y clientes comparten un solo lugar para
saber dónde va cada entrega y en qué estado está.
```

```
</p>
```

```
<p class="pie-marca">Sistema Logístico de El Salvador</p>
```

```
</section>
```

```
<section class="panel-formulario">
```

```
<div class="tarjeta-login">
```

```
<div class="selector-rol" id="login-roles">
```

```
<button type="button" class="rol-btn activo"
```

```
onclick="setLoginRole('admin')">Administrador</button>
```

```
<button type="button" class="rol-btn"
```

```
onclick="setLoginRole('conductor')">Conductor</button>
```

```
<button type="button" class="rol-btn" onclick="setLoginRole('cliente')">Cliente</button>
```

```
</div>
```

```
<h2 class="form-titulo">Inicia sesión</h2>
```

<p class="form-subtitulo">Ingresá tus credenciales para continuar.</p>

<form id="loginForm" class="form-login">

<label class="campo">

Usuario

<input type="text" id="login_user" placeholder="Ej: admin" required value="admin" />

</label>

<label class="campo">

Contraseña

<input type="password" id="login_pass" placeholder="••••••••" required value="123" />

</label>

<button type="submit" class="btn-entrar">Entrar al Sistema</button>

</form>

</div>

</section>

</main>

<!-- ===== APLICACIÓN PRINCIPAL ===== -->

<div id="app-screen">

<header>

<div class="header-branding">

<svg viewBox="0 0 32 32" width="18" height="18">

<path d="M4 20 C 9 8, 18 22, 28 6" fill="none" stroke="#2ec4b6" stroke-width="2.6" stroke-linecap="round"/>

<circle cx="4" cy="20" r="3" fill="#f5a623"/>

<circle cx="28" cy="6" r="3" fill="#f5a623"/>

</svg>

Transporte & Entregas

</div>

<div class="user-info">

Usuario

ADMIN

<button class="btn-salir" onclick="logout()">Salir</button>

</div>

</header>

<div class="main-container">

<!-- Navegación Lateral -->

<nav class="sidebar-nav">

<button id="nav-pedidos" class="active" onclick="switchTab('pedidos')> Pedidos y
Rutas</button>

<button id="nav-clientes" class="admin-only" onclick="switchTab('clientes')>
Clientes</button>

<button id="nav-conductores" class="admin-only" onclick="switchTab('conductores')>
Conductores</button>

<button id="nav-vehiculos" class="admin-only" onclick="switchTab('vehiculos')>
Vehículos</button>

</nav>

<!-- Panel de Formularios -->

<aside class="panel-form" id="panelForm">

<!-- Búsqueda Nominatim -->

<div class="tarjeta" id="card-search">

<h3>Geolocalizar Dirección</h3>

<p class="sub">Buscá un punto en El Salvador para centrar en el mapa.</p>

<div class="campo" style="gap:8px;">

<input type="text" id="search-address" placeholder="Ej: San Salvador, Santa Tecla">

<button class="btn btn-ambar" type="button" onclick="searchAddress()">Buscar

Ubicación</button>

</div>

</div>

<!-- Formulario CRUD Pedidos -->

<div class="tarjeta crud-form" id="form-pedidos">

<h3 id="title-pedidos">Registrar Pedido</h3>

<p class="sub">Completa los datos del envío y ubicación.</p>

<form id="orderForm" style="display:flex; flex-direction:column; gap:12px;">

<input type="hidden" id="id_pedido">

<div class="campo admin-only">

Cliente

<select id="fk_cliente"></select>

</div>

<div class="campo admin-only">

Asignar Conductor

<select id="fk_conductor"></select>

</div>

<div class="campo admin-only">

Asignar Vehículo

<select id="fk_vehiculo"></select>

</div>

<div class="campo">

Dirección de Entrega

<input type="text" id="direccion" placeholder="Selecciona un punto en el mapa" required>

</div>

<div style="display: flex; gap: 8px;">

<div class="campo" style="flex: 1;">

Latitud

<input type="text" id="latitud" readonly required>

</div>

<div class="campo" style="flex: 1;">

Longitud

```
<input type="text" id="longitud" readonly required>
</div>
</div>
```

```
<div class="campo admin-only">
  <span>Estado Inicial</span>
  <select id="estado_admin">
    <option value="PENDIENTE">PENDIENTE</option>
    <option value="ASIGNADO">ASIGNADO</option>
    <option value="EN_CAMINO">EN_CAMINO</option>
    <option value="ENTREGADO">ENTREGADO</option>
    <option value="INCIDENCIA">INCIDENCIA</option>
    <option value="CANCELADO">CANCELADO</option>
  </select>
</div>
```

```
<button type="submit" id="btn-save-pedido" class="btn btn-primario">Guardar
Pedido</button>
<button type="button" id="btn-cancel-pedido" class="btn btn-secundario oculto"
onclick="resetOrderForm()">Cancelar</button>
</form>
</div>
```

```
<!-- Formulario CRUD Clientes -->
<div class="tarjeta crud-form oculto" id="form-clientes">
  <h3 id="title-clientes">Agregar Cliente</h3>
  <p class="sub">Registra la información del nuevo cliente.</p>
  <form id="clientForm" style="display:flex; flex-direction:column; gap:12px;">
    <input type="hidden" id="id_cliente">
    <div class="campo"><span>Nombre Completo</span><input type="text" id="cli_nombre"
required></div>
    <div class="campo"><span>Teléfono</span><input type="text" id="cli_telefono"
required></div>
    <div class="campo"><span>Email</span><input type="email" id="cli_email" required></div>
    <div class="campo"><span>Dirección Fiscal</span><input type="text" id="cli_direccion"
required></div>
    <button type="submit" class="btn btn-primario">Guardar Cliente</button>
    <button type="button" id="btn-cancel-cliente" class="btn btn-secundario oculto"
onclick="resetClientForm()">Cancelar</button>
  </form>
</div>
```

```
<!-- Formulario CRUD Conductores -->
<div class="tarjeta crud-form oculto" id="form-conductores">
  <h3 id="title-conductores">Agregar Conductor</h3>
  <p class="sub">Datos del conductor y su licencia.</p>
  <form id="driverForm" style="display:flex; flex-direction:column; gap:12px;">
    <input type="hidden" id="id_conductor">
```

```

        <div class="campo"><span>Nombre Completo</span><input type="text" id="cond_nombre"
required></div>
        <div class="campo"><span>Nº Licencia</span><input type="text" id="cond_licencia"
required></div>
        <div class="campo"><span>Teléfono</span><input type="text" id="cond_telefono"
required></div>
        <button type="submit" class="btn btn-primario">Guardar Conductor</button>
        <button type="button" id="btn-cancel-conductor" class="btn btn-secundario oculto"
onclick="resetDriverForm()">Cancelar</button>
    </form>
</div>

```

```

<!-- Formulario CRUD Vehículos -->
<div class="tarjeta crud-form oculto" id="form-vehiculos">
    <h3 id="title-vehiculos">Agregar Vehículo</h3>
    <p class="sub">Registro de unidad de transporte.</p>
    <form id="vehicleForm" style="display:flex; flex-direction:column; gap:12px;">
        <input type="hidden" id="id_vehiculo">
        <div class="campo"><span>Placa</span><input type="text" id="veh_placa" required></div>
        <div class="campo"><span>Tipo de Vehículo</span><input type="text" id="veh_tipo"
placeholder="Ej: Camión 3.5T" required></div>
        <div class="campo"><span>Capacidad</span><input type="text" id="veh_capacidad"
placeholder="Ej: 1200 kg" required></div>
        <button type="submit" class="btn btn-primario">Guardar Vehículo</button>
        <button type="button" id="btn-cancel-vehiculo" class="btn btn-secundario oculto"
onclick="resetVehicleForm()">Cancelar</button>
    </form>
</div>
</aside>

```

```

<!-- Área Principal de Mapa y Tabla -->
<main class="content-area">
    <div id="map"></div>
    <div class="table-container">
        <div class="table-top">
            <h3 id="table-title" class="table-title">Lista de Pedidos</h3>
        </div>
        <table class="tabla">
            <thead id="table-head"></thead>
            <tbody id="table-body"></tbody>
        </table>
    </div>
</main>
</div>
</div>

```

```

<script>
    const API_BASE_URL = "http://localhost:8000";

```

```

// Variables globales
let currentUser = null;
let currentTab = 'pedidos';
let savedMarkers = {};

// Caché local en memoria
let clientesDB = [];
let conductoresDB = [];
let vehiculosDB = [];
let pedidosDB = [];

function setLoginRole(rol) {
  document.querySelectorAll('#login-roles .rol-btn').forEach(btn => btn.classList.remove('activo'));
  event.target.classList.add('activo');
  if(rol === 'admin') document.getElementById('login_user').value = 'admin';
  if(rol === 'conductor') document.getElementById('login_user').value = 'rsilva';
  if(rol === 'cliente') document.getElementById('login_user').value = 'jperez';
}

// Reglas de negocio para estados
const SystemRules = {
  validTransitions: {
    'PENDIENTE': ['ASIGNADO', 'CANCELADO'],
    'ASIGNADO': ['EN_CAMINO', 'PENDIENTE', 'CANCELADO'],
    'EN_CAMINO': ['ENTREGADO', 'INCIDENCIA'],
    'INCIDENCIA': ['EN_CAMINO', 'CANCELADO', 'ENTREGADO'],
    'ENTREGADO': [],
    'CANCELADO': []
  },
  validateTransition(order, newStatus, user) {
    const currentStatus = order.estado;
    if (currentStatus === newStatus) return { allowed: true };

    if (!this.validTransitions[currentStatus]?.includes(newStatus)) {
      return { allowed: false, reason: `Transición no permitida: No se puede cambiar de '$
{currentStatus}' a '${newStatus}'.` };
    }

    if (currentStatus === 'ENTREGADO' || currentStatus === 'CANCELADO') {
      return { allowed: false, reason: `El pedido está en estado final (${currentStatus}).` };
    }

    if (user.rol === 'CLIENTE' && newStatus === 'CANCELADO' && currentStatus !==
'PENDIENTE') {
      return { allowed: false, reason: 'No puedes cancelar un pedido asignado o en proceso.' };
    }

    if (user.rol === 'CONDUCTOR' && order.id_conductor !== user.id_ref) {
      return { allowed: false, reason: 'Solo puedes cambiar el estado de tus propios pedidos.' };
    }
  }
};

```

```

    }

    if (newStatus === 'EN_CAMINO' && (!order.id_conductor || !order.id_vehiculo)) {
        return { allowed: false, reason: 'Se requiere asignar Conductor y Vehículo antes de pasar a
EN_CAMINO.' };
    }

    return { allowed: true };
}
};

// Autenticación
document.getElementById('loginForm').addEventListener('submit', async function(e) {
    e.preventDefault();
    const u = document.getElementById('login_user').value;
    const p = document.getElementById('login_pass').value;

    try {
        const res = await fetch(`${API_BASE_URL}/login`, {
            method: 'POST',
            headers: { 'Content-Type': 'application/json' },
            body: JSON.stringify({ username: u, password: p })
        });

        if (!res.ok) {
            alert("Credenciales incorrectas");
            return;
        }

        currentUser = await res.json();
        document.getElementById('login-screen').style.display = 'none';
        document.getElementById('app-screen').style.display = 'flex';
        document.getElementById('user-display-name').innerText = currentUser.nombre;
        document.getElementById('user-role-badge').innerText = currentUser.rol;

        applyPermissions();
        await loadAllData();
        switchTab('pedidos');
    } catch (err) {
        console.error(err);
        alert("Error al conectar con la API");
    }
});

function logout() {
    currentUser = null;
    document.getElementById('app-screen').style.display = 'none';
    document.getElementById('login-screen').style.display = 'grid';
}

```

```
function applyPermissions() {
  document.querySelectorAll('.admin-only').forEach(el => {
    el.style.display = (currentUser.rol === 'ADMIN') ? '' : 'none';
  });

  if (currentUser.rol === 'CONDUCTOR') {
    document.getElementById('card-search').style.display = 'none';
    document.getElementById('form-pedidos').style.display = 'none';
  } else {
    document.getElementById('card-search').style.display = 'block';
    document.getElementById('form-pedidos').style.display = 'block';
  }
}
```

// Carga de Datos

```
async function loadAllData() {
  try {
    const [resCli, resCond, resVeh, resPed] = await Promise.all([
      fetch(`${API_BASE_URL}/clientes`),
      fetch(`${API_BASE_URL}/conductores`),
      fetch(`${API_BASE_URL}/vehiculos`),
      fetch(`${API_BASE_URL}/pedidos`)
    ]);
```

```
    clientesDB = await resCli.json();
    conductoresDB = await resCond.json();
    vehiculosDB = await resVeh.json();
    pedidosDB = await resPed.json();
```

```
    populateSelects();
  } catch (err) {
    console.error("Error al cargar datos:", err);
  }
}
```

```
function populateSelects() {
  document.getElementById('fk_cliente').innerHTML = clientesDB.map(c => `<option value="${c.id_cliente}">${c.nombre}</option>`).join("");
  document.getElementById('fk_conductor').innerHTML = '<option value="">-- Sin Asignar --</option>' + conductoresDB.map(c => `<option value="${c.id_conductor}">${c.nombre}</option>`).join("");
  document.getElementById('fk_vehiculo').innerHTML = '<option value="">-- Sin Asignar --</option>' + vehiculosDB.map(v => `<option value="${v.id_vehiculo}">${v.tipo} (${v.placa})</option>`).join("");
}
```

// Mapa MapLibre GL

```
const elSalvadorBounds = [[-90.1500, 13.1000], [-87.6500, 14.4500]];
```

```

const defaultCoords = [-89.2182, 13.6929];

const map = new maplibregl.Map({
  container: 'map',
  style: {
    version: 8,
    sources: { 'osm-tiles': { type: 'raster', tiles: ['https://tile.openstreetmap.org/{z}/{x}/{y}.png'],
tileSize: 256 } },
    layers: [{ id: 'osm-layer', type: 'raster', source: 'osm-tiles' }]
  },
  center: defaultCoords, zoom: 12, maxBounds: elSalvadorBounds, renderWorldCopies: false
});

map.on('load', () => map.resize());
map.addControl(new maplibregl.NavigationControl(), 'top-right');

const activeMarker = new maplibregl.Marker({ color: '#f5a623', draggable: true })
  .setLngLat(defaultCoords).addTo(map);

activeMarker.on('dragend', () => {
  const { lat, lng } = activeMarker.getLngLat();
  updateCoordinatesInput(lat, lng);
  reverseGeocode(lat, lng);
});

map.on('click', (e) => {
  if (currentUser?.rol === 'CONDUCTOR') return;
  const { lat, lng } = e.lngLat;
  activeMarker.setLngLat([lng, lat]);
  updateCoordinatesInput(lat, lng);
  reverseGeocode(lat, lng);
});

function updateCoordinatesInput(lat, lng) {
  document.getElementById('latitud').value = parseFloat(lat).toFixed(6);
  document.getElementById('longitud').value = parseFloat(lng).toFixed(6);
}
updateCoordinatesInput(defaultCoords[1], defaultCoords[0]);

async function searchAddress() {
  const query = document.getElementById('search-address').value;
  if (!query) return;
  const url = `https://nominatim.openstreetmap.org/search?format=json&q=${
encodeURIComponent(query)}&countrycodes=sv`;
  try {
    const res = await fetch(url);
    const data = await res.json();
    if (data && data.length > 0) {
      const lat = parseFloat(data[0].lat), lon = parseFloat(data[0].lon);

```

```

    map.flyTo({ center: [lon, lat], zoom: 14 });
    activeMarker.setLngLat([lon, lat]);
    updateCoordinatesInput(lat, lon);
    document.getElementById('direccion').value = data[0].display_name;
  } else {
    alert('Ubicación no encontrada.');
```

```
  }
```

```
} catch (err) { console.error(err); }
```

```
}
```

```
async function reverseGeocode(lat, lon) {
```

```
  const url = `https://nominatim.openstreetmap.org/reverse?format=json&lat=${lat}&lon=${lon}`;
```

```
  try {
```

```
    const res = await fetch(url);
```

```
    const data = await res.json();
```

```
    if (data && data.display_name) document.getElementById('direccion').value = data.display_name;
```

```
  } catch (err) { console.error(err); }
```

```
}
```

```
// Navegación
```

```
function switchTab(tab) {
```

```
  currentTab = tab;
```

```
  document.querySelectorAll('.sidebar-nav button').forEach(b => b.classList.remove('active'));
```

```
  document.getElementById(`nav-${tab}`).classList.add('active');
```

```
  document.querySelectorAll('.crud-form').forEach(f => f.classList.add('oculto'));
```

```
  if (tab === 'pedidos') {
```

```
    document.getElementById('map').style.display = 'block';
```

```
    if (currentUser.rol !== 'CONDUCTOR') document.getElementById('form-pedidos').classList.remove('oculto');
```

```
  } else {
```

```
    document.getElementById('map').style.display = 'none';
```

```
    document.getElementById(`form-${tab}`).classList.remove('oculto');
```

```
  }
```

```
  map.resize();
```

```
  renderCurrentView();
```

```
}
```

```
function renderCurrentView() {
```

```
  const thead = document.getElementById('table-head');
```

```
  const tbody = document.getElementById('table-body');
```

```
  tbody.innerHTML = "";
```

```
  Object.keys(savedMarkers).forEach(k => savedMarkers[k].remove());
```

```
  savedMarkers = {};
```

```
  if (currentTab === 'pedidos') {
```

```
    document.getElementById('table-title').innerText = `Gestión de Pedidos (${currentUser.rol})`;
```

```

thead.innerHTML =
`<tr><th>ID</th><th>Cliente</th><th>Conductor</th><th>Vehículo</th><th>Dirección</th><th>Estado</th><th>Acciones</th></tr>`;

let filtered = pedidosDB;
if (currentUser.rol === 'CONDUCTOR') filtered = pedidosDB.filter(p => p.id_conductor === currentUser.id_ref);
if (currentUser.rol === 'CLIENTE') filtered = pedidosDB.filter(p => p.id_cliente === currentUser.id_ref);

filtered.forEach(p => {
  const cli = clientesDB.find(c => c.id_cliente === p.id_cliente)?.nombre || 'N/A';
  const cond = conductoresDB.find(c => c.id_conductor === p.id_conductor)?.nombre || 'Sin Asignar';
  const veh = vehiculosDB.find(v => v.id_vehiculo === p.id_vehiculo)?.placa || 'Sin Asignar';

  let actions = "";
  if (currentUser.rol === 'ADMIN') {
    actions = `<button class="btn-action btn-edit"
onclick="editOrder(${p.id_pedido})">Editar</button>
<button class="btn-action btn-del"
onclick="deleteOrder(${p.id_pedido})">Eliminar</button>`;
  } else if (currentUser.rol === 'CONDUCTOR') {
    actions = `
<select style="font-size:0.75rem; padding:4px;" onchange="changeOrderStatus(${p.id_pedido}, this.value)">
<option value="ASIGNADO" ${p.estado==='ASIGNADO'?selected:''}>Asignado</option>
<option value="EN_CAMINO" ${p.estado==='EN_CAMINO'?selected:''}>En camino</option>
<option value="ENTREGADO"
${p.estado==='ENTREGADO'?selected:''}>Entregado</option>
<option value="INCIDENCIA"
${p.estado==='INCIDENCIA'?selected:''}>Incidencia</option>
</select>`;
  } else if (currentUser.rol === 'CLIENTE') {
    actions = `
    ${p.estado === 'PENDIENTE' ? `<button class="btn-action btn-edit" onclick="editOrder(${p.id_pedido})">Editar</button>` : ""}
    ${p.estado === 'PENDIENTE' ? `<button class="btn-action btn-del"
onclick="changeOrderStatus(${p.id_pedido}, 'CANCELADO')">Cancelar</button>` : `<span
style="font-size:0.75rem; color:gray;">En proceso</span>`}
  `;
  }

tbody.innerHTML += `<tr>
<td><b>#${p.id_pedido}</b></td><td>${cli}</td><td>${cond}</td><td>${veh}</td>
<td title="${p.direccion}">${p.direccion.substring(0, 24)}...</td>
<td><span class="badge badge-${p.estado}">${p.estado}</span></td>
<td>${actions}</td>

```

```

</tr>`;

const m = new maplibregl.Marker({ color: getMarkerColor(p.estado) })
    .setLngLat([p.longitud, p.latitud])
    .setPopup(new maplibregl.Popup().setHTML(`<b>Pedido #${p.id_pedido}</b><br>Estado: $
{p.estado}<br>Cliente: ${cli}`))
    .addTo(map);
savedMarkers[p.id_pedido] = m;
});

} else if (currentTab === 'clientes') {
    document.getElementById('table-title').innerText = 'CRUD Clientes';
    thead.innerHTML =
`<tr><th>ID</th><th>Nombre</th><th>Teléfono</th><th>Email</th><th>Dirección</
th><th>Acciones</th></tr>`;
    clientesDB.forEach(c => {
        tbody.innerHTML += `<tr>
            <td>${c.id_cliente}</td><td>${c.nombre}</td><td>${c.telefono}</td><td>${c.email}</
td><td>${c.direccion}</td>
            <td>
                <button class="btn-action btn-edit" onclick="editClient(${c.id_cliente})">Editar</button>
                <button class="btn-action btn-del" onclick="deleteClient(${c.id_cliente})">Eliminar</button>
            </td></tr>`;
    });
} else if (currentTab === 'conductores') {
    document.getElementById('table-title').innerText = 'CRUD Conductores';
    thead.innerHTML =
`<tr><th>ID</th><th>Nombre</th><th>Licencia</th><th>Teléfono</th><th>Acciones</th></tr>`;
    conductoresDB.forEach(c => {
        tbody.innerHTML += `<tr>
            <td>${c.id_conductor}</td><td>${c.nombre}</td><td>${c.licencia}</td><td>${c.telefono}
</td>
            <td>
                <button class="btn-action btn-edit" onclick="editDriver(${c.id_conductor})">Editar</button>
                <button class="btn-action btn-del"
onclick="deleteDriver(${c.id_conductor})">Eliminar</button>
            </td></tr>`;
    });
} else if (currentTab === 'vehiculos') {
    document.getElementById('table-title').innerText = 'CRUD Vehículos';
    thead.innerHTML =
`<tr><th>ID</th><th>Placa</th><th>Tipo</th><th>Capacidad</th><th>Acciones</th></tr>`;
    vehiculosDB.forEach(v => {
        tbody.innerHTML += `<tr>
            <td>${v.id_vehiculo}</td><td>${v.placa}</td><td>${v.tipo}</td><td>${v.capacidad}</td>
            <td>
                <button class="btn-action btn-edit" onclick="editVehicle(${v.id_vehiculo})">Editar</button>
                <button class="btn-action btn-del"
onclick="deleteVehicle(${v.id_vehiculo})">Eliminar</button>
            </td></tr>`;
    });
}

```

```

        </td></tr>`;
    });
}
}

```

```

function getMarkerColor(st) {
  switch(st) {
    case 'PENDIENTE': return '#f5a623';
    case 'ASIGNADO': return '#805ad5';
    case 'EN_CAMINO': return '#3182ce';
    case 'ENTREGADO': return '#2ec4b6';
    case 'INCIDENCIA': return '#ef4b4b';
    default: return '#718096';
  }
}

```

// CRUD APIs

```

async function changeOrderStatus(orderId, newStatus) {
  const order = pedidosDB.find(p => p.id_pedido === orderId);
  if (!order) return;

  const val = SystemRules.validateTransition(order, newStatus, currentUser);
  if (!val.allowed) {
    alert(`Acción Denegada: ${val.reason}`);
    renderCurrentView();
    return;
  }
}

```

```

try {
  const payload = { ...order, estado: newStatus };
  const res = await fetch(`${API_BASE_URL}/pedidos/${orderId}`, {
    method: 'PUT',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify(payload)
  });
  if (res.ok) {
    await loadAllData();
    renderCurrentView();
  }
} catch (err) { console.error(err); }
}

```

// CRUD Pedidos

```

document.getElementById('orderForm').addEventListener('submit', async function(e) {
  e.preventDefault();
  const id = document.getElementById('id_pedido').value;
  const cId = (currentUser.rol === 'CLIENTE') ? currentUser.id_ref :
parseInt(document.getElementById('fk_cliente').value);
  const condVal = document.getElementById('fk_conductor').value;

```

```

const vehVal = document.getElementById('fk_vehiculo').value;

let targetStatus = 'PENDIENTE';
if (currentUser.rol === 'ADMIN') {
  targetStatus = document.getElementById('estado_admin').value;
  if (targetStatus === 'PENDIENTE' && condVal && vehVal) targetStatus = 'ASIGNADO';
}

const bodyData = {
  id_cliente: cId,
  id_conductor: condVal ? parseInt(condVal) : null,
  id_vehiculo: vehVal ? parseInt(vehVal) : null,
  direccion: document.getElementById('direccion').value,
  latitud: parseFloat(document.getElementById('latitud').value),
  longitud: parseFloat(document.getElementById('longitud').value),
  estado: targetStatus
};

try {
  let res;
  if (id) {
    res = await fetch(`${API_BASE_URL}/pedidos/${id}`, {
      method: 'PUT',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(bodyData)
    });
  } else {
    res = await fetch(`${API_BASE_URL}/pedidos`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(bodyData)
    });
  }

  if (res.ok) {
    resetOrderForm();
    await loadAllData();
    renderCurrentView();
  } else {
    alert('Error guardando el pedido');
  }
} catch (err) { console.error(err); }
});

function editOrder(id) {
  const p = pedidosDB.find(x => x.id_pedido === id);
  if (!p) return;
  document.getElementById('id_pedido').value = p.id_pedido;
  document.getElementById('fk_cliente').value = p.id_cliente;
}

```

```

document.getElementById('fk_conductor').value = p.id_conductor || "";
document.getElementById('fk_vehiculo').value = p.id_vehiculo || "";
document.getElementById('direccion').value = p.direccion;
document.getElementById('latitud').value = p.latitud;
document.getElementById('longitud').value = p.longitud;
document.getElementById('estado_admin').value = p.estado;

activeMarker.setLngLat([p.longitud, p.latitud]);
map.flyTo({ center: [p.longitud, p.latitud], zoom: 14 });

document.getElementById('title-pedidos').innerText = `Editar Pedido #${p.id_pedido}`;
document.getElementById('btn-save-pedido').innerText = 'Actualizar Pedido';
document.getElementById('btn-cancel-pedido').classList.remove('oculto');
}

```

```

async function deleteOrder(id) {
  if (!confirm('¿Eliminar este pedido?')) return;
  await fetch(`${API_BASE_URL}/pedidos/${id}`, { method: 'DELETE' });
  await loadAllData();
  renderCurrentView();
}

```

```

function resetOrderForm() {
  document.getElementById('orderForm').reset();
  document.getElementById('id_pedido').value = "";
  document.getElementById('title-pedidos').innerText = 'Registrar Pedido';
  document.getElementById('btn-save-pedido').innerText = 'Guardar Pedido';
  document.getElementById('btn-cancel-pedido').classList.add('oculto');
  updateCoordinatesInput(defaultCoords[1], defaultCoords[0]);
}

```

// CRUD Clientes

```

document.getElementById('clientForm').addEventListener('submit', async function(e) {
  e.preventDefault();
  const id = document.getElementById('id_cliente').value;
  const bodyData = {
    nombre: document.getElementById('cli_nombre').value,
    telefono: document.getElementById('cli_telefono').value,
    email: document.getElementById('cli_email').value,
    direccion: document.getElementById('cli_direccion').value
  };

```

```

const url = id ? `${API_BASE_URL}/clientes/${id}` : `${API_BASE_URL}/clientes`;
const method = id ? 'PUT' : 'POST';

```

```

await fetch(url, {
  method: method,
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify(bodyData)

```

```

});

resetClientForm();
await loadAllData();
renderCurrentView();
});

function editClient(id) {
  const c = clientesDB.find(x => x.id_cliente === id);
  if (!c) return;
  document.getElementById('id_cliente').value = c.id_cliente;
  document.getElementById('cli_nombre').value = c.nombre;
  document.getElementById('cli_telefono').value = c.telefono;
  document.getElementById('cli_email').value = c.email;
  document.getElementById('cli_direccion').value = c.direccion;
  document.getElementById('title-clientes').innerText = `Editar Cliente #${id}`;
  document.getElementById('btn-cancel-cliente').classList.remove('oculto');
}

async function deleteClient(id) {
  if (!confirm('¿Eliminar cliente?')) return;
  await fetch(`${API_BASE_URL}/clientes/${id}`, { method: 'DELETE' });
  await loadAllData();
  renderCurrentView();
}

function resetClientForm() {
  document.getElementById('clientForm').reset();
  document.getElementById('id_cliente').value = "";
  document.getElementById('title-clientes').innerText = 'Agregar Cliente';
  document.getElementById('btn-cancel-cliente').classList.add('oculto');
}

// CRUD Conductores
document.getElementById('driverForm').addEventListener('submit', async function(e) {
  e.preventDefault();
  const id = document.getElementById('id_conductor').value;
  const bodyData = {
    nombre: document.getElementById('cond_nombre').value,
    licencia: document.getElementById('cond_licencia').value,
    telefono: document.getElementById('cond_telefono').value
  };

  const url = id ? `${API_BASE_URL}/conductores/${id}` : `${API_BASE_URL}/conductores`;
  const method = id ? 'PUT' : 'POST';

  await fetch(url, {
    method: method,
    headers: { 'Content-Type': 'application/json' },

```

```
body: JSON.stringify(bodyData)
});
```

```
resetDriverForm();
await loadAllData();
renderCurrentView();
});
```

```
function editDriver(id) {
  const d = conductoresDB.find(x => x.id_conductor === id);
  if (!d) return;
  document.getElementById('id_conductor').value = d.id_conductor;
  document.getElementById('cond_nombre').value = d.nombre;
  document.getElementById('cond_licencia').value = d.licencia;
  document.getElementById('cond_telefono').value = d.telefono;
  document.getElementById('title-conductores').innerText = `Editar Conductor #${id}`;
  document.getElementById('btn-cancel-conductor').classList.remove('oculto');
}
```

```
async function deleteDriver(id) {
  if (!confirm('¿Eliminar conductor?')) return;
  await fetch(`${API_BASE_URL}/conductores/${id}`, { method: 'DELETE' });
  await loadAllData();
  renderCurrentView();
}
```

```
function resetDriverForm() {
  document.getElementById('driverForm').reset();
  document.getElementById('id_conductor').value = "";
  document.getElementById('title-conductores').innerText = 'Agregar Conductor';
  document.getElementById('btn-cancel-conductor').classList.add('oculto');
}
```

// CRUD Vehículos

```
document.getElementById('vehicleForm').addEventListener('submit', async function(e) {
  e.preventDefault();
  const id = document.getElementById('id_vehiculo').value;
  const bodyData = {
    placa: document.getElementById('veh_placa').value,
    tipo: document.getElementById('veh_tipo').value,
    capacidad: document.getElementById('veh_capacidad').value
  };
});
```

```
const url = id ? `${API_BASE_URL}/vehiculos/${id}` : `${API_BASE_URL}/vehiculos`;
const method = id ? 'PUT' : 'POST';
```

```
await fetch(url, {
  method: method,
  headers: { 'Content-Type': 'application/json' },
```

```

    body: JSON.stringify(bodyData)
  });

  resetVehicleForm();
  await loadAllData();
  renderCurrentView();
});

function editVehicle(id) {
  const v = vehiculosDB.find(x => x.id_vehiculo === id);
  if (!v) return;
  document.getElementById('id_vehiculo').value = v.id_vehiculo;
  document.getElementById('veh_placa').value = v.placa;
  document.getElementById('veh_tipo').value = v.tipo;
  document.getElementById('veh_capacidad').value = v.capacidad;
  document.getElementById('title-vehiculos').innerText = `Editar Vehículo #${id}`;
  document.getElementById('btn-cancel-vehiculo').classList.remove('oculto');
}

async function deleteVehicle(id) {
  if (!confirm('¿Eliminar vehículo?')) return;
  await fetch(`${API_BASE_URL}/vehiculos/${id}`, { method: 'DELETE' });
  await loadAllData();
  renderCurrentView();
}

function resetVehicleForm() {
  document.getElementById('vehicleForm').reset();
  document.getElementById('id_vehiculo').value = "";
  document.getElementById('title-vehiculos').innerText = 'Agregar Vehículo';
  document.getElementById('btn-cancel-vehiculo').classList.add('oculto');
}
</script>
</body>
</html>

```